Fertility, Mortality, Caregiving, And Lifecycle Labor Allocation

Timing, Persistence, Retirement, And Late-Life Work

Special Topic 6: Labor Market, Health, and Population – Week 3

Central question

- How do fertility timing, mortality risk, aging, and caregiving burdens reshape labor allocation over the life cycle?
- The labor objects are hours, participation, earnings, occupation, mobility, savings, retirement timing, bridge jobs, and welfare.
- Week 3 discipline: current shocks can move future wages, occupations, savings, retirement, and household specialization.

Why this is labor economics

- Fertility affects work because births arrive at specific career stages and change time allocation.
- Severe health shocks and mortality risk change household resources, care needs, insurance value, and expected horizons.
- Caregiving changes feasible hours, jobs, mobility, retirement timing, and demand for flexibility.
- Aging societies make pension rules, older-worker job design, rehiring, and post-retirement work central labor-market margins.

Lifecycle labor-allocation framework

$$V_t(s_t) = \max_{h_t, c_t, j_t, a_{t+1}, r_t} u(C_t, L_t, q_t, \ell_t) + \beta E_t[V_{t+1}(s_{t+1})]$$

- States: age, health, children, spouse and parent health, assets, pension eligibility, job type, accumulated experience.
- Choices: market work, care time, occupation, mobility, savings, retirement, bridge work, post-retirement work.
- Shocks: birth, severe illness, death, caregiving onset, pension eligibility, health decline, employer flexibility.
- Dynamic point: today's hours, care, and job choices alter tomorrow's wages, options, savings, and welfare.

Fertility timing and child penalties

Object	Labor channel	Dynamic consequence
Birth timing	leave, hours, childcare, job continuity	experience path
Child penalty	gendered earnings and hours divergence	persistent gaps
Occupation choice	flexibility, travel, task demands	wage growth tradeoff
Promotion timing	tenure and ladder interruption	slower advancement
Specialization	household market-care split	future outside options

- Separate timing effects from persistent specialization and career-path effects.

Family responses to severe health shocks

Shock channel	Household response	Identification caution
Income loss	spouse or adult child works more	anticipation
Care demand	family member works less or switches jobs	hidden intensity
Mortality risk	savings, insurance, retirement plans change	expected horizon
Health capacity	own work declines after illness	severity
Liquidity stress	short-run hours and asset responses	insurance access

- The sign of family labor supply is ambiguous because income replacement and care time push in opposite directions.

Caregiving and household reallocation

- Informal care is a labor-market margin: it changes hours, participation, job choice, mobility, and retirement.
- Parent care, spouse care, and elder care differ in predictability, intensity, co-residence, distance, and formal-care substitutes.
- Paid leave and formal care can preserve attachment by lowering the time cost of family health shocks.
- Hidden margins matter: forgone overtime, declined promotion, reduced travel, lower search, and earlier claiming may be missed.

Aging, pensions, retirement, and post-retirement work

Margin	What moves	Labor interpretation
Pension eligibility	benefit claiming and exit timing	incentive effect
Retirement savings	ability to reduce work or absorb shocks	insurance margin
Health decline	capacity and job feasibility	work constraint
Flexible work	hours, tasks, schedules, remote work	job-design margin
Bridge job / rehire	partial exit, return, part-time work	retirement transition

- Retirement is often a transition into a different labor arrangement, not simply an absorbing exit.

Global aging evidence

- Cross-country evidence helps separate health, preferences, pension rules, employment protection, and care systems.
- Population aging changes labor-force attachment, retirement timing, care demand, older-worker job design, and firm adaptation.
- The labor question is not demographic drag by itself; it is which work, wage, care, productivity, and retirement margins adjust.
- Week 5 will scale up to aggregate demographic change; Week 3 keeps the lifecycle household and retirement margins visible.

Research Lab design

- Primary anchor: Fadlon and Nielsen on family labor-supply responses to severe health shocks.
- Challenge anchor: Ameriks and coauthors on older-worker flexibility and late-life job design.
- Reproduce: estimate a synthetic family health-shock event profile.
- Diagnose: classify income replacement, care demand, work-capacity loss, pension incentives, flexibility, and selection.
- Transfer: redesign the logic for child penalties, caregiving onset, pension reform, or bridge-job transitions.

Research design checklist

1. Name the event: birth, illness, death, care onset, pension eligibility, aging threshold, or retirement transition.
2. Name the margin: hours, participation, occupation, savings, retirement, rehire, bridge job, or welfare.
3. State the horizon: immediate disruption, medium-run reallocation, long-run career path, or late-life adjustment.
4. Separate income replacement, care demand, work capacity, pension incentives, flexibility, and selection.
5. Say what remains invisible in earnings or employment.

Bridge to Week 4

- Week 3 centers household timing, care burdens, aging, pensions, and retirement transitions.
- Week 4 moves inside workplaces: mental health, stress, productivity, absenteeism, presenteeism, retention, and worker welfare.
- The common thread is dynamic labor adjustment after health and demographic shocks.

Takeaways

- Fertility, mortality, caregiving, and aging are labor-market events because they change work, wages, job choice, savings, and welfare.
- Timing and persistence are central: a current shock can alter future career and retirement states.
- Caregiving and late-life flexibility are core allocation margins, not side topics.
- Global aging matters when it changes institutions and job design around older-worker labor supply and care demand.